

## EC2 & Web hosting using Windows instance

### Class no:13

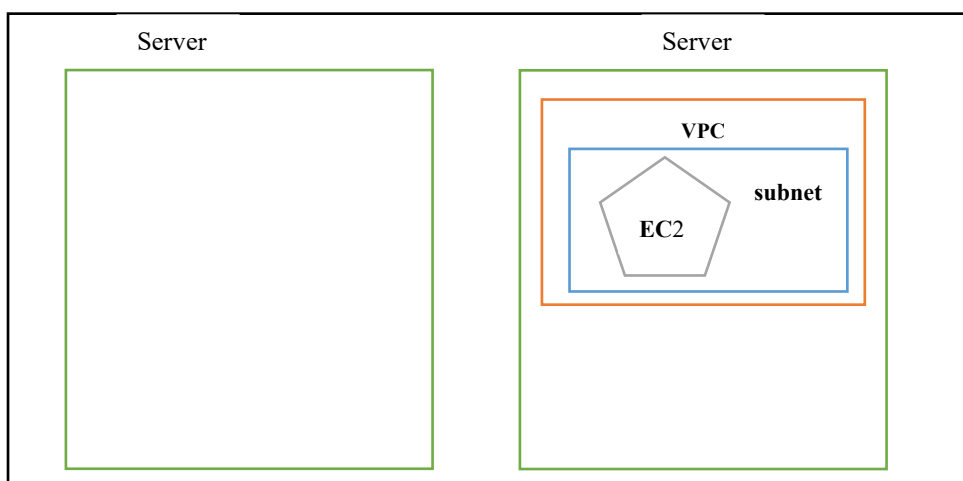
### EC2:

Amazon EC2 (Elastic Compute Cloud) is AWS's core service for renting virtual servers in the cloud. To configure EC2, you need to create an instance by choosing an AMI (Amazon Machine Image), instance type, storage, networking, and security settings. EC2 instance families are divided into categories like General Purpose, Compute Optimized, Memory Optimized, Storage Optimized, and Accelerated Computing, each balancing performance and price differently.

**To configure an instance, we need:**

- Name
- OS
- Instance type -> CPU & Memory
- Key pair (Public and Private key)
- Disk Storage (Persistence Volume)
- Network settings
  - ✓ VPC (Virtual Private Cloud)-larger network
  - ✓ Subnet (smaller network)
  - ✓ Auto-assign public Ip ( Enabled: public ec2/ Disabled: private ec2)
  - ✓ Security groups
  - ✓ Disk Storage (Persistence Volume)

**Layout of an Ec2 instance:**



### INSTANCE TYPE

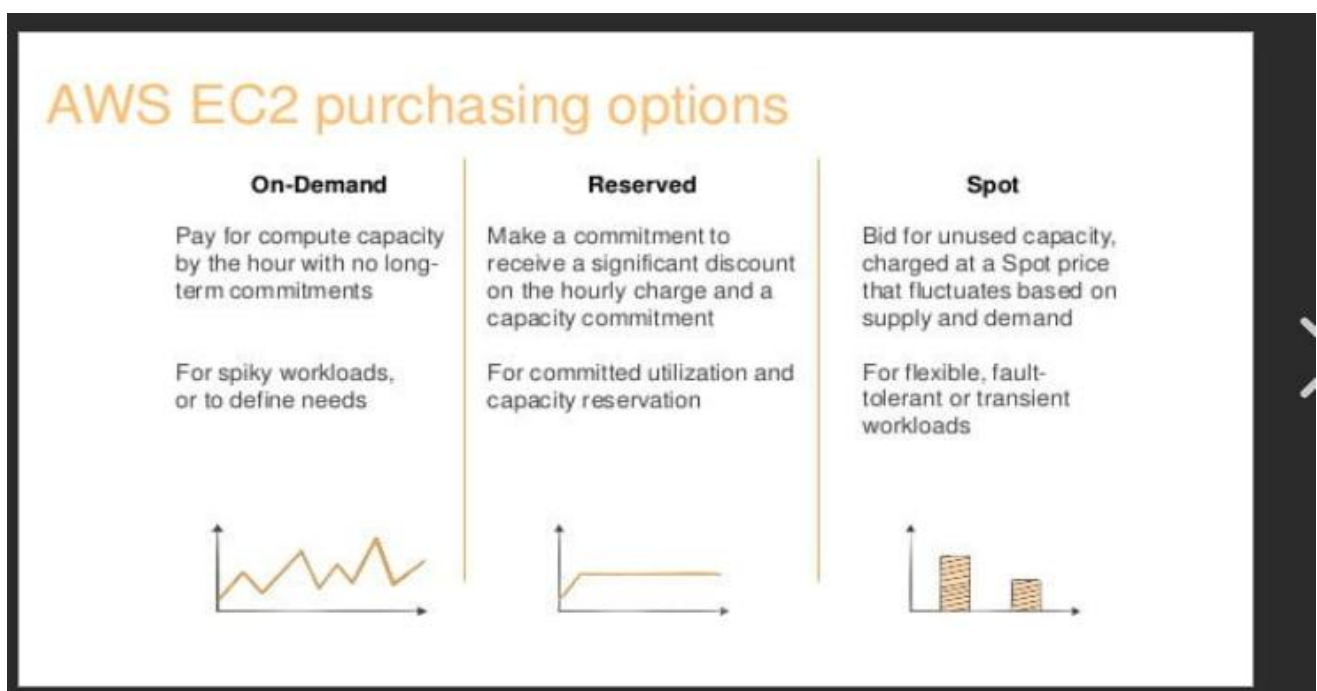
Amazon EC2 instance types are predefined virtual server configurations that combine varying amounts of compute, memory, storage, and networking capacity to suit different workloads. Choosing the right type ensures optimal performance and cost-efficiency.

## Main Categories: (Performance based)

- **General Purpose** – Balanced compute, memory, and networking. Ideal for web servers, code repositories, and small-to-medium databases. Examples: T / M Series
- **Compute Optimized** – High-performance processors for compute-bound tasks like batch processing, media transcoding, and game servers. Examples: C series
- **Memory Optimized** – High memory-to-CPU ratio for large in-memory datasets, analytics, and enterprise apps. Examples: R / X / Z Series
- **Accelerated Computing** – Use GPUs or FPGAs for graphics rendering, ML training, or scientific simulations. Examples: P5, G6e, Inf2, Trn1.
- **Storage Optimized** – High sequential read/write and IOPS for large datasets. Examples: I / D /H Series.
- **High-Performance Computing (HPC)** – Optimized for large-scale simulations and deep learning. Examples: Hpc7g, Hpc6a.

## Purchasing or Pricing models

- On demand: On-Demand Instances provide **predictable pricing** and are billed per second with no long-term commitments. These instances are ideal for workloads that require **reliable and uninterrupted performance**.
- Spot instance: Spot Instances allow you to utilize unused EC2 capacity at significantly reduced costs, often up to **90% cheaper** than On-Demand pricing. However, these instances come with the risk of **interruption** when AWS reclaims capacity for On-Demand or Reserved Instances. Spot Instances are ideal for workloads that are **flexible** and can tolerate interruptions.
- Reserved instance: Reserved Instances (RIs) are a **billing discount model** offered by AWS for services like Amazon EC2 and Amazon RDS. They allow users to save significantly—up to **75% compared to On-Demand pricing**—by committing to a specific instance configuration for a one-year or three-year term. Unlike physical instances, RIs are not actual servers but rather a **pricing mechanism** applied to On-Demand Instances that match specific attributes.



## KEY PAIR:

- key pair is a set of public and private keys used for secure login to your EC2 instances.
- AWS uses public-key cryptography:
  - (i)The public key is stored in AWS and attached to your EC2 instance.
  - (ii)The private key is downloaded by you as a .pem file when you create the key pair.
- When you connect via SSH (Linux) or RDP (Windows), AWS checks that your private key matches the public key stored on the instance.
- The public key is injected into your EC2 instance at launch:
  - For **Linux instances**, it's placed inside the file: `.ssh/authorized_keys`.
  - Use command: `ls -la ->` To see the hidden files.

## Hands on Web hosting using windows EC2 instance:

**Step no:1** Create a Windows Ec2 instance and enable Https in the Network Settings. Launch instance.

### Name and tags [Info](#)

Name

[Add additional tags](#)

### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)  
Including AMIs from AWS, Marketplace and the Community

#### Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base  
ami-0f23ee11d840e3e64 (64-bit (x86))  
Virtualization: hvm    ENA enabled: true    Root device type: ebs

Free tier eligible

#### Description

Microsoft Windows Server 2025 Datacenter edition. [English]

Microsoft Windows Server 2025 Full Locale English AMI provided by Amazon

| Architecture | AMI ID                | Publish Date | Username      |
|--------------|-----------------------|--------------|---------------|
| 64-bit (x86) | ami-0f23ee11d840e3e64 | 2026-03-12   | Administrator |

Verified provider

### ▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro  
Family: t3    2 vCPU    1 GiB Memory    Current generation: true    On-Demand Linux base pricing: 0.0108 USD per Hour  
On-Demand Windows base pricing: 0.02 USD per Hour    On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour  
On-Demand RHEL base pricing: 0.0346 USD per Hour    On-Demand SUSE base pricing: 0.0108 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

### ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

batchfrw

▼

[Create new key pair](#)

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

## ▼ Network settings [Info](#)

[Edit](#)

### Network [Info](#)

vpc-08d23a3e5bc324ead

### Subnet [Info](#)

No preference (Default subnet in any availability zone)

### Auto-assign public IP [Info](#)

Enable

### Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-9' with the following rules:

☒ Allow RDP traffic from  
Helps you connect to your instance

Anywhere  
0.0.0.0/0

☐ Allow HTTPS traffic from the internet  
To set up an endpoint, for example when creating a web server

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⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

## ▼ Configure storage [Info](#)

[Advanced](#)

1x 30 GiB gp3 Root volume, 3000 IOPS, Not encrypted

**Step no:2** Connect the windows instance. Go to the RDP client.

## Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID  
i-Oec19453e0e301b46 (My windows machine)

VPC ID  
vpc-08d23a3e5bc324ead

Security groups  
sg-0484935f754cd1232 (launch-wizard-9)

IAM role  
-

SSM Session Manager [RDP client](#) [EC2 serial console](#)

① Record RDP connections  
You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#)

Try for free [i](#) ×

Instance ID  
i-Oec19453e0e301b46 (My windows machine)

### Connection Type

☒ Connect using RDP client  
Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager  
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS  
ec2-13-61-16-158.eu-north-1.compute.amazonaws.com

### Username [Info](#)

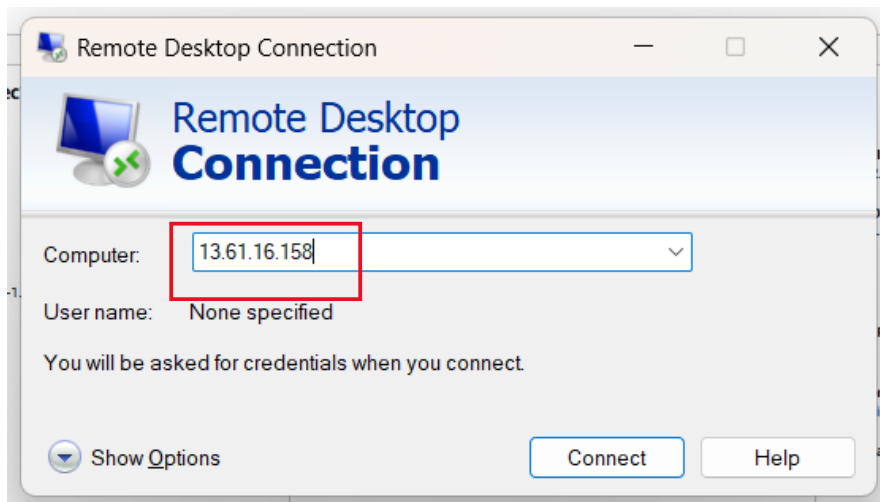
[i](#) Administrator

Password [Get password](#)

② If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

[Cancel](#)

**Step no:3** Open Remote desktop connection in our local machine and open. Paste the public ip address of the ec2 instance. Click connect.



Get the password by uploading the keypair file and decrypt it.

**Get Windows password** [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password.

**Instance ID**  
i-Dec19453e0e301b46 (My windows machine)

**Key pair associated with this instance**  
batchfrw

**Private key**  
Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

batchfrw.pem  
1.67 KB

**Private key contents**

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEA4zxeDIYwAppt35w0DgWqePJPF0ousNPLuIiZuBXBdfcIM82
B8rQIPDJGO9D/DGjus+QUPvgll/IR6UOyRrZkM+kOHILbGD3SKn3viTPi4pk/5Ne
lx2uL0sb4LrE7FbC6VD3thbw037dfouCBw8CP5/LFqQv7J29t2/JcEdwhMNTQl4
GkJe0IWu5jBYccP2RDsiQxzcdyhQD6SXwcojrw2UitInlTq3r15j6ZlDfRDgOG
RE/WgV/H2yheIRy4VsxQK0PcnaGOCV/Ma0ZuM5nvcX2OmnhRHkGyEZ/rR2zkBjwj
B85TmbOAO98byG0a+6RnAYmc3HMqVyhTn3WkiwiDAQABaolBAGmkMwXvQ7nws
Dq1GLQ61d9eYAUTr9lCxCyXTRq5jI8QdkoBHBnN92/ELJMcMP0o19C+dyBMxc7
-----
```

**Password**  
AfP&IzK7rjIKSHjDgzzwbd22ePn1;Qlt

[If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.](#)

Enter the password in the RDC and click connect.

**Windows Security**

**Enter your credentials**

These credentials will be used to connect to 13.61.16.158.

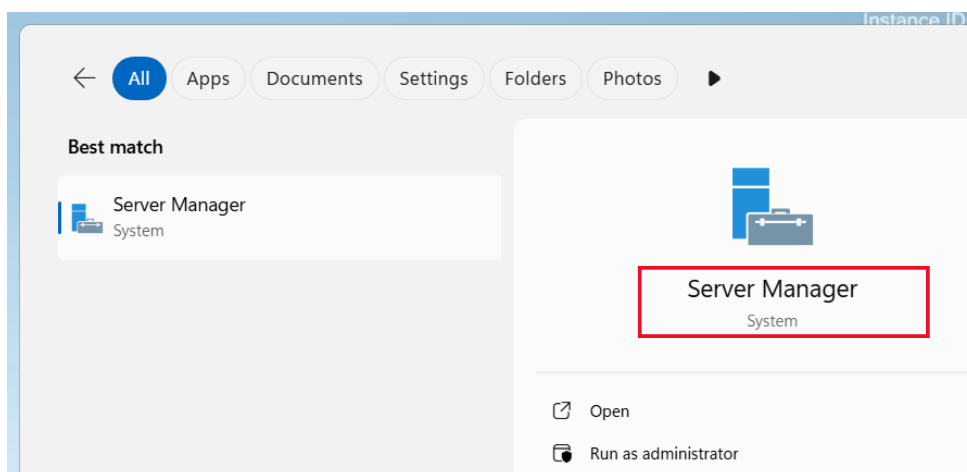
**User name**  
Administrator

**Password**  
[Password field with dots and an eye icon]

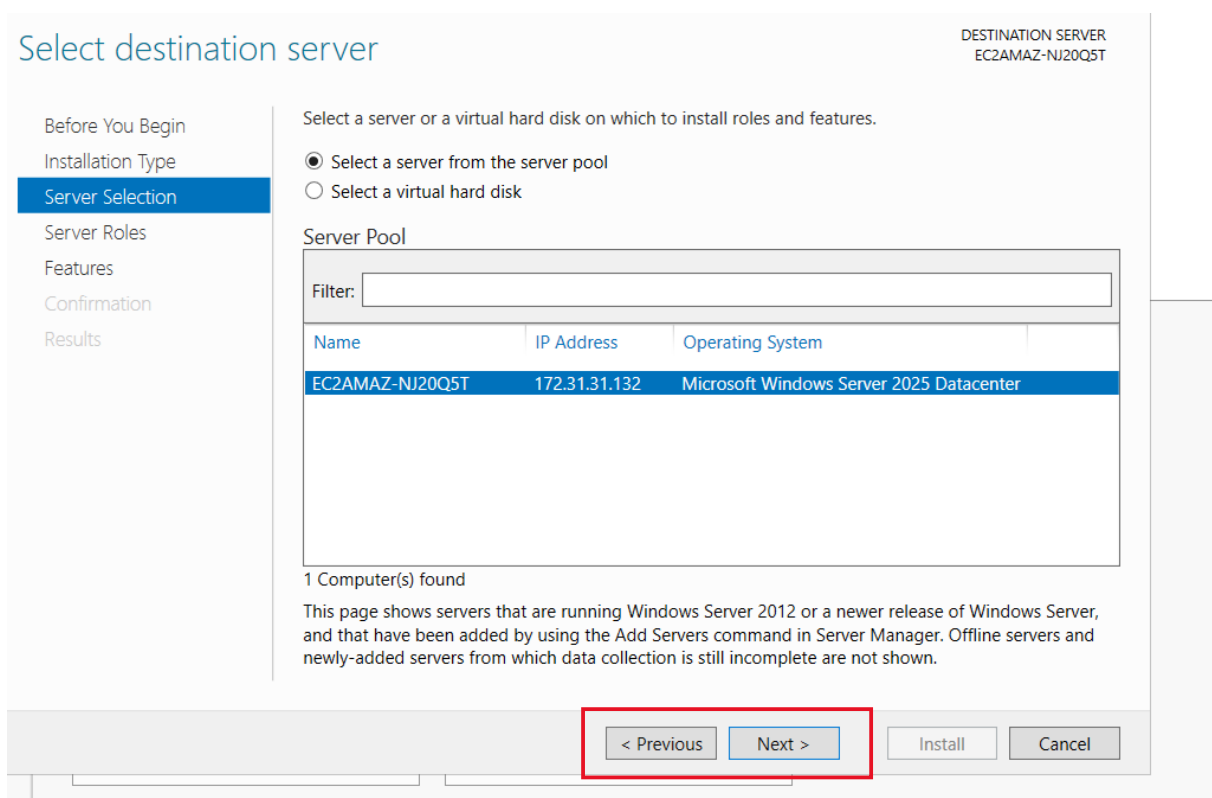
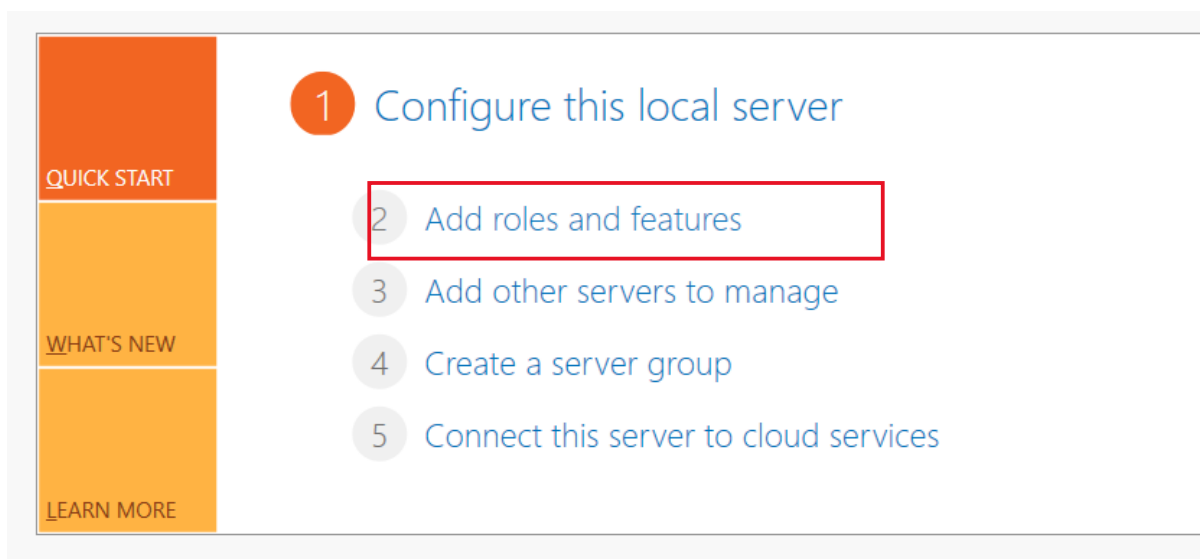
☐ Remember me

**OK** **Cancel**

**Step no:4** Now the windows machine will open. Search server manager and open it.



**Step no:5** Click on add roles and features. Click on next until server roles.



**Step no:6** Click we server (IIS) and choose add features.

Server Selection

Server Roles

Features

Confirmation

Results

- ☐ Active Directory Domain Services
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☐ DNS Server
- ☐ Fax Server
- ☒ File and Storage Services (1 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ Network Controller
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☒ Web Server (IIS)
- ☐ Windows Deployment Services
- ☐ Windows Server Update Services

Windows Server 2016

Add Roles and Features Wizard

Add features that are required for Web Server (IIS)?

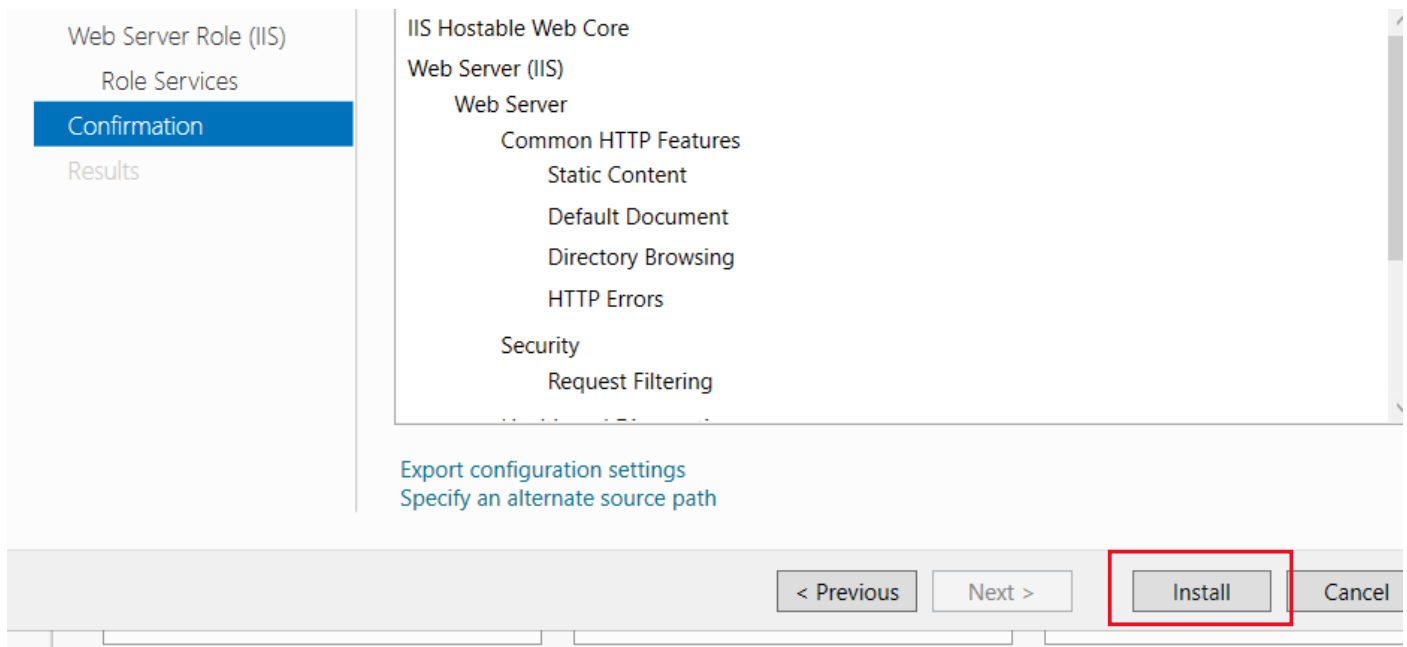
The following tools are required to manage this feature, but do not have to be installed on the same server.

- Web Server (IIS)
  - Management Tools
    - [Tools] IIS Management Console

☒ Include management tools (if applicable)

Add Features Cancel

**Step no:7**Now click install.



**Step no:8** Now copy the public ip address of the ec2 instance and paste it in an incognito browser we can see the output.

#### Instance summary for i-0ec19453e0e301b46 (My windows machine) [Info](#)

Updated less than a minute ago

##### Instance ID

[i-0ec19453e0e301b46](#)

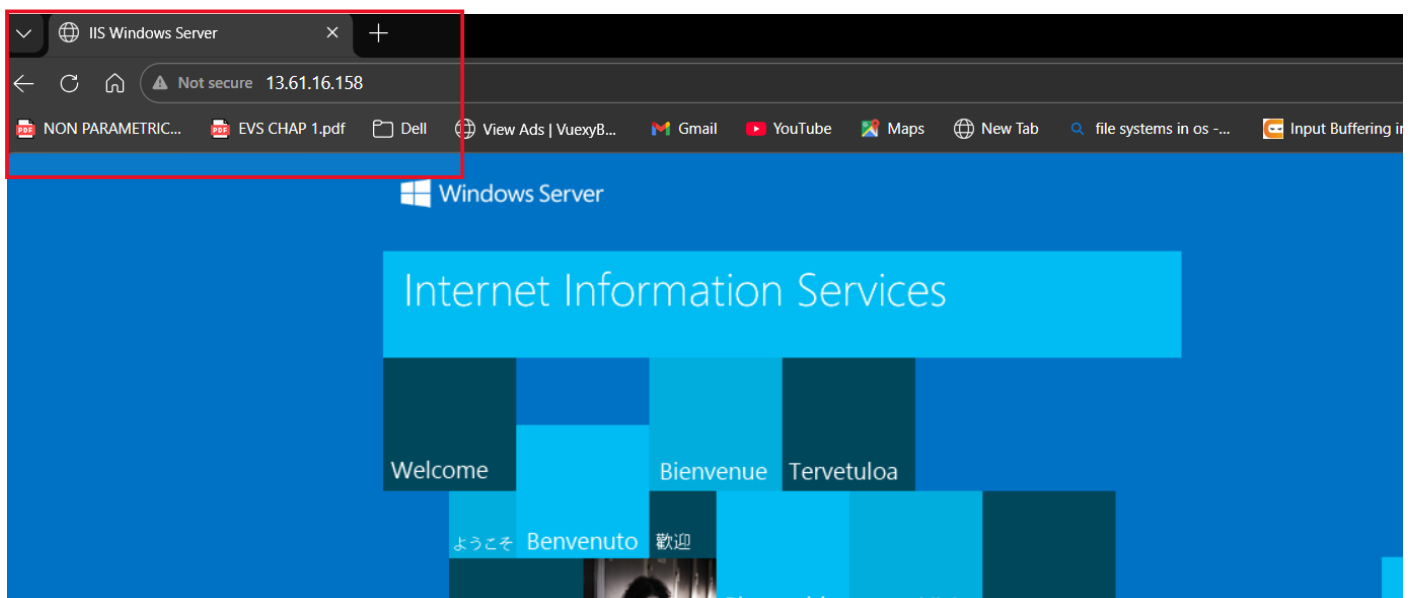
##### IPv6 address

##### Public IPv4 address

[13.61.16.158](#) | [open address](#)

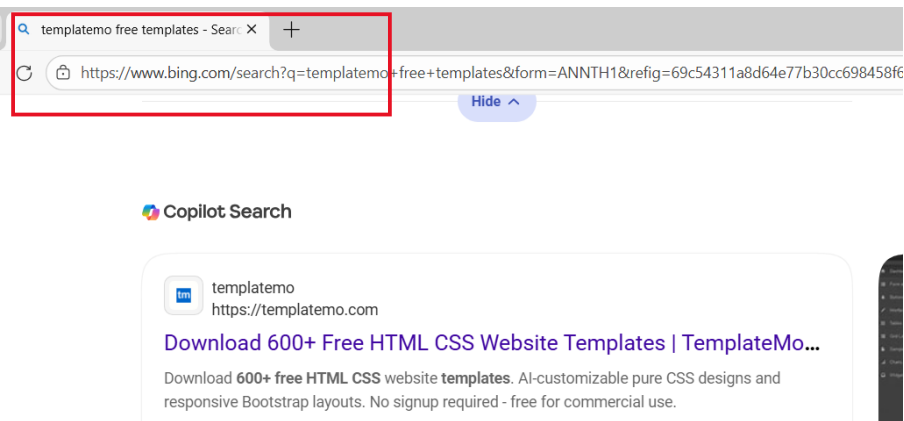
##### Instance state

[View state](#)

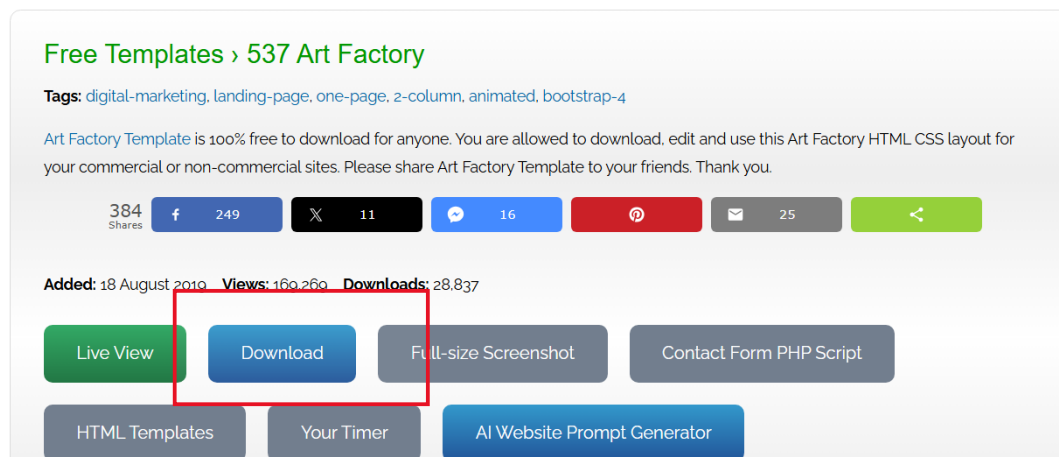


**Step no:9** Now we can add our file and we can host it. Go to our Windows instance. Open the edge browser , search for free templates.

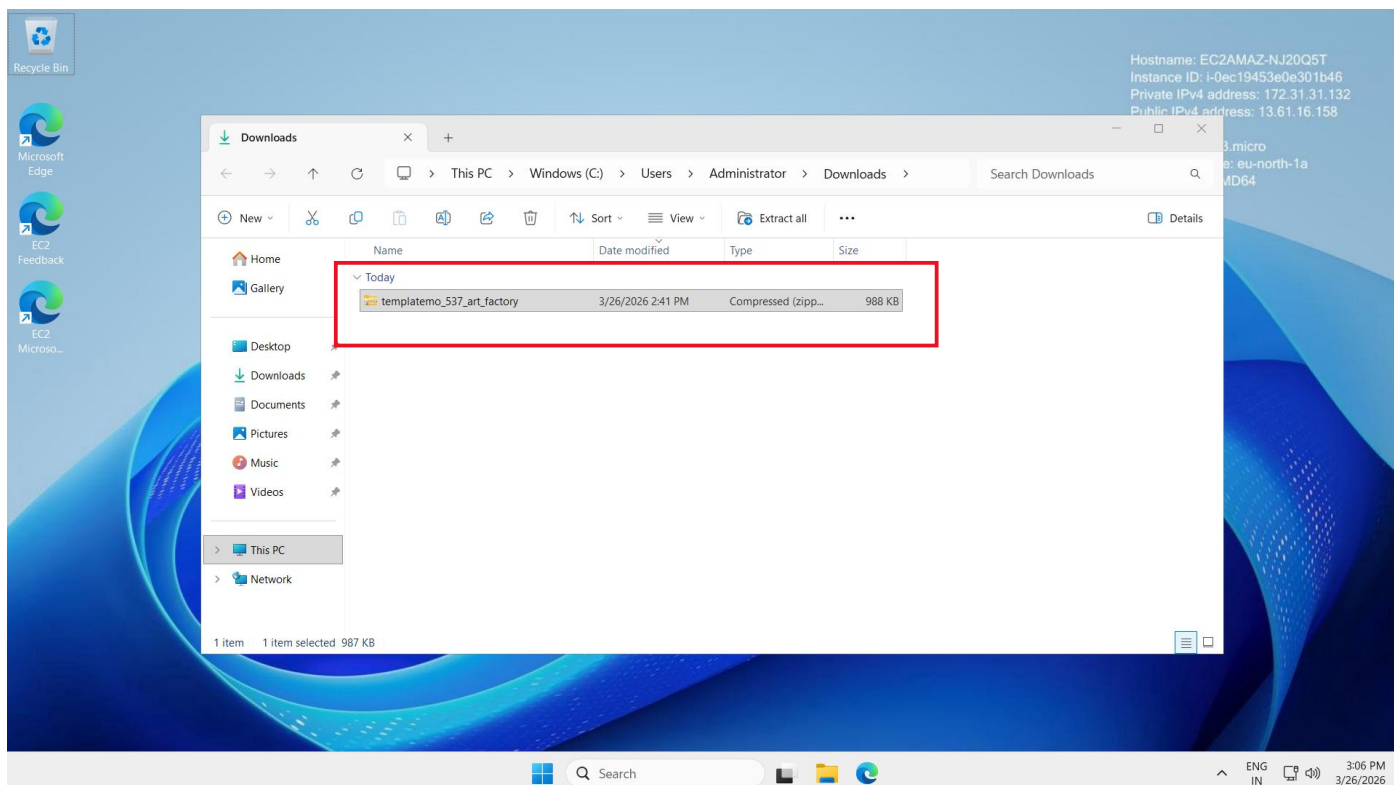




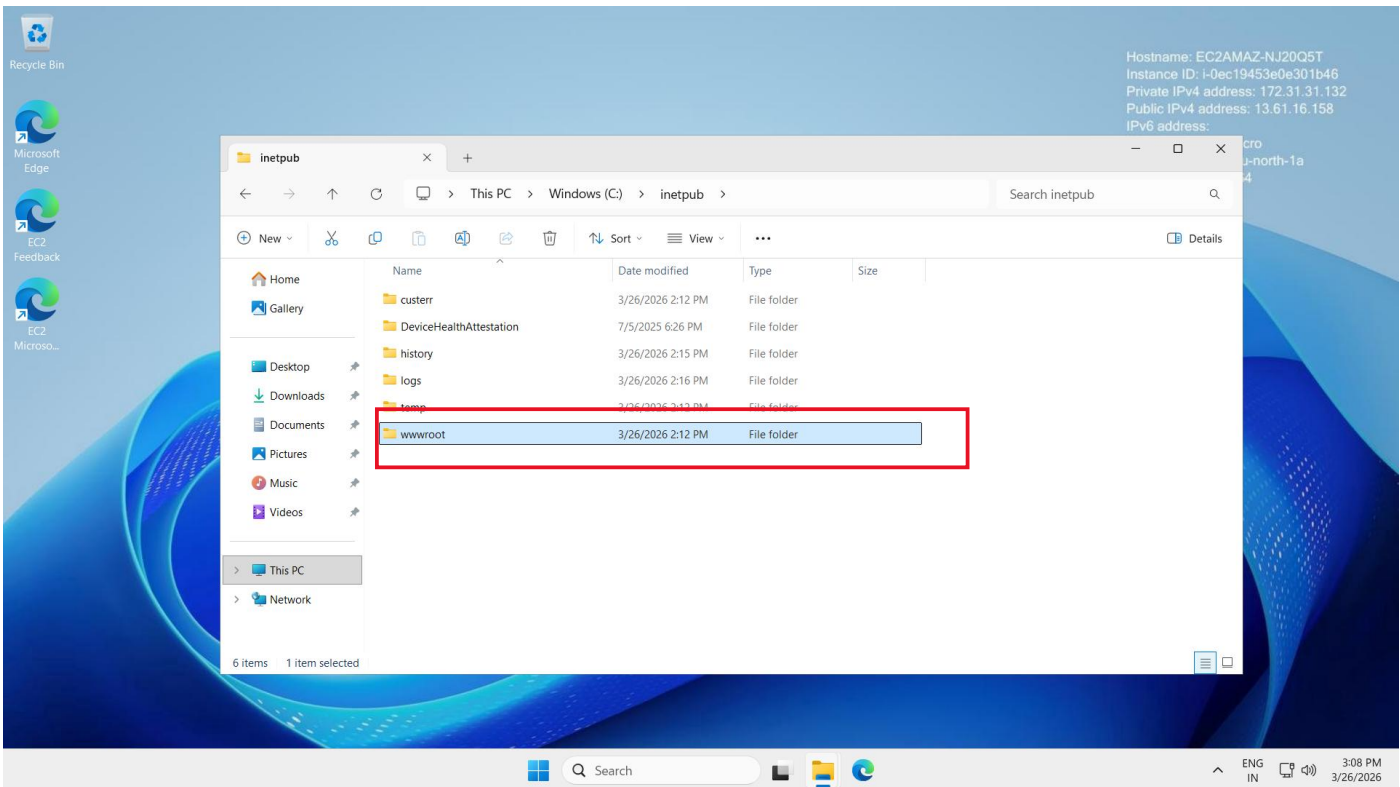
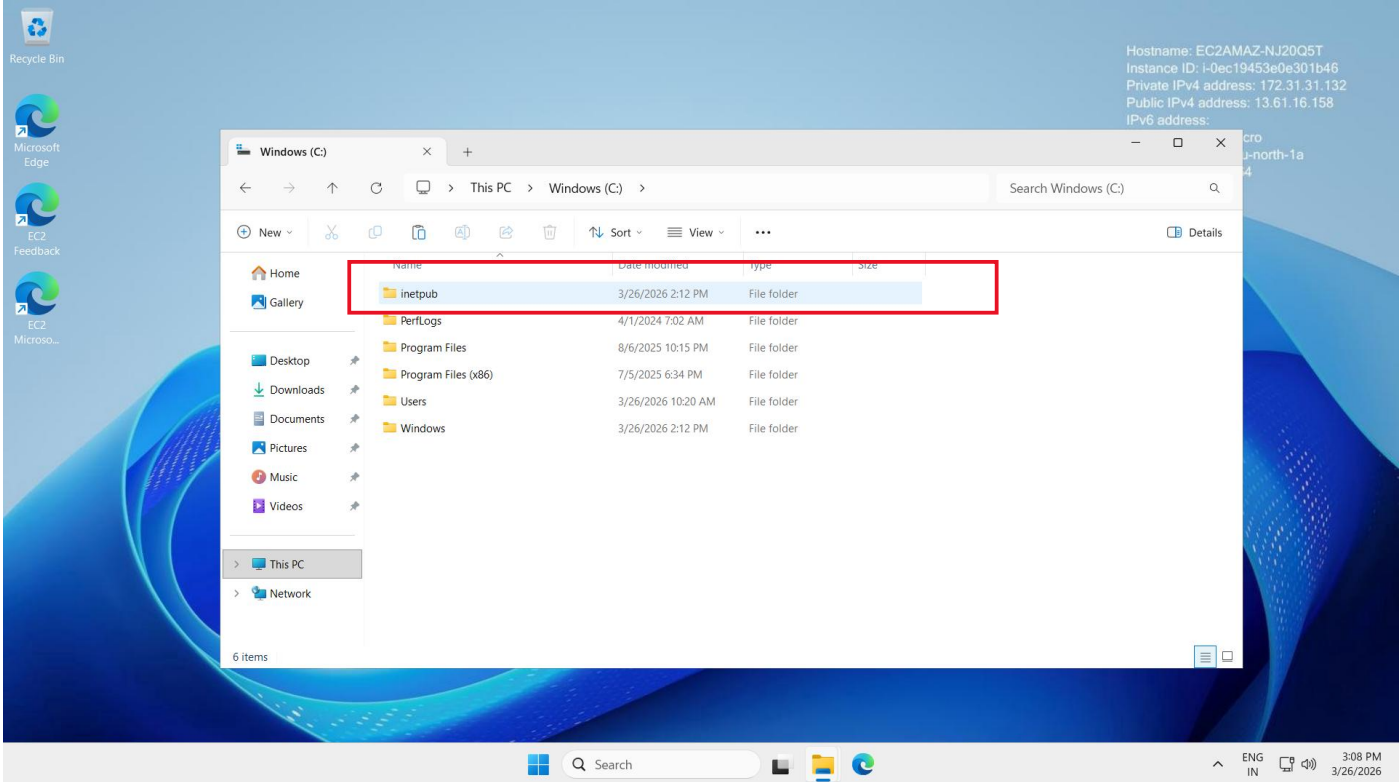
**Step no:10** Now open it download it

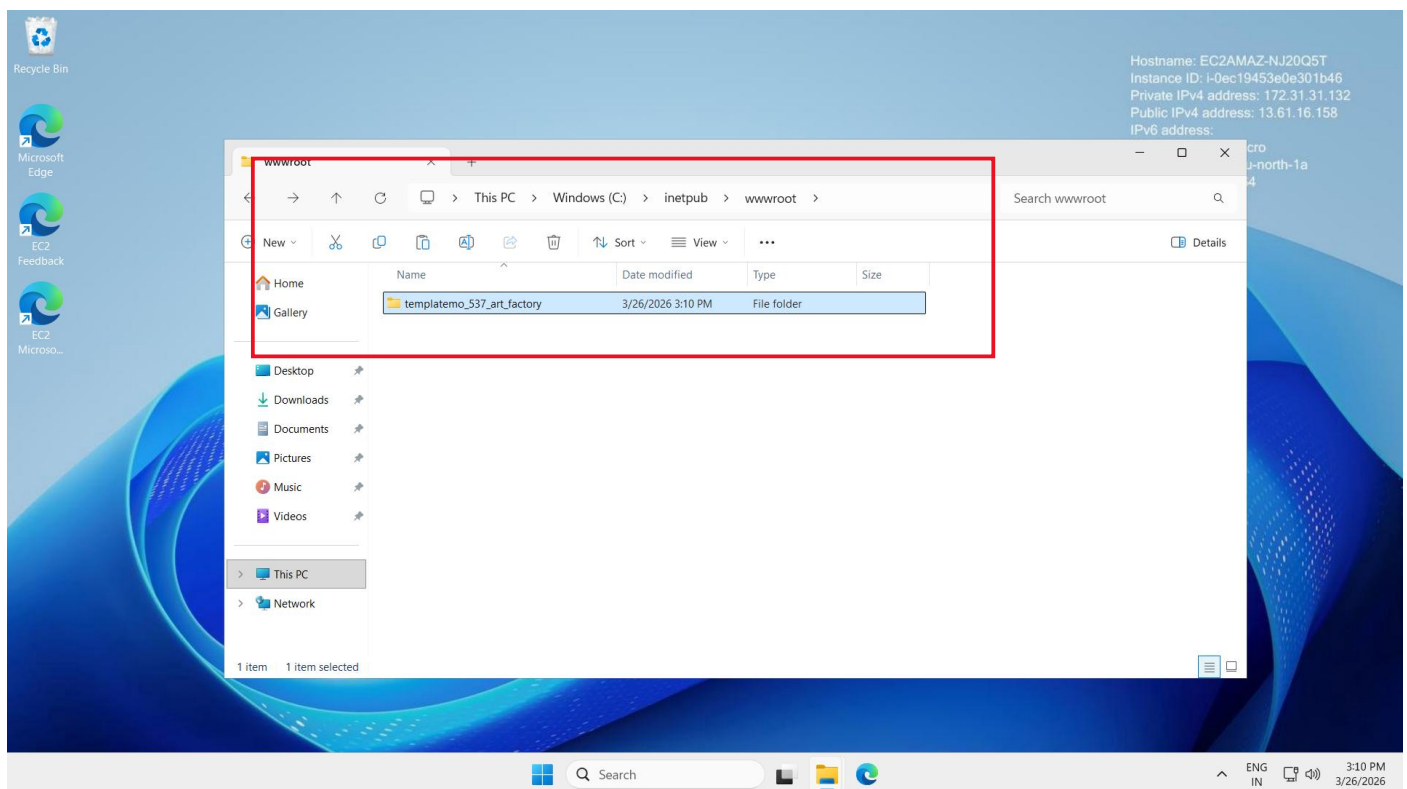
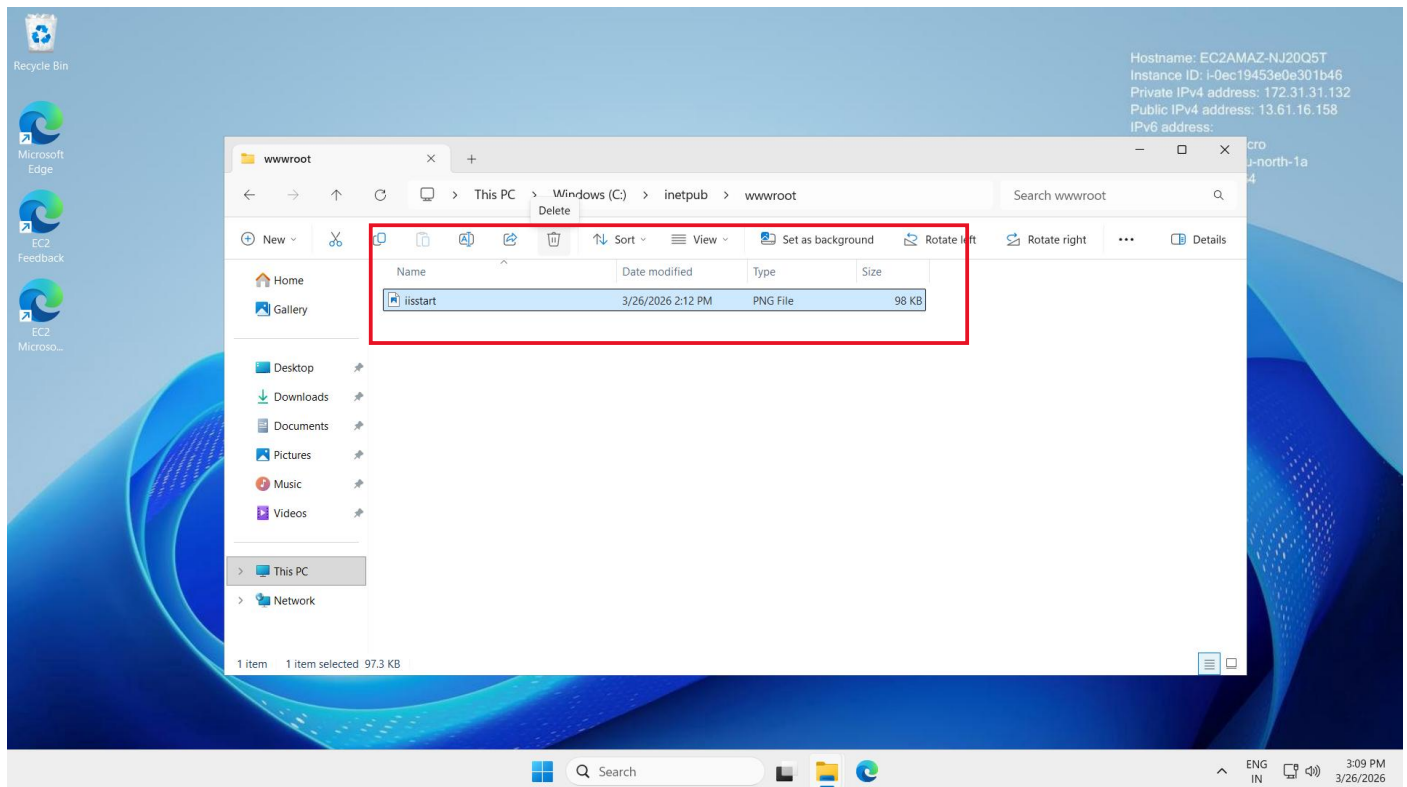


**Step no:11** Now open the file in the downloads. It will be a zipfile. Extract it



**Step no:12** Now follow the path →this pc→ inetpub → wwwroot → delete the existing html file and paste our downloaded file.





Now copy the public ip of the ec2 and paste it in a private browser. We can see the output of our code.